Badri Narayanan Murali Krishnan

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EXPERIENCE

Engineer, AI Software Tools **Mar 2026 – Present**
Qualcomm *San Diego, CA*
- Working to design and **optimize** high-performance **ML and GenAI models** on **ONNX Runtime**.
- Applying **advanced optimization techniques**, including **model quantization** and **graph-level transformations**, to ensure efficient execution and high system reliability on power-constrained hardware.

AI / ML Engineer **Sep 2025 – Feb 2026**
CuroNow *Madison, WI*
- **Cross-collaborated** to design a **Document Extraction AI Agent** with upload, ingestion, & extraction APIs, including monitoring/observability.
- Architected a **Medication Agent** to provide reliable medication insights using **LLM reasoning**, **fallback handling**.
- Built a **centralized AI caching microservice** that reduced duplicate LLM API costs by **40%** and latency by **25%**.

AI / ML Engineer - Contract Role **Jun 2025 – Sep 2025**
BrainWaves Digital *Remote, USA*
- **Led the development** of a **Finance Agent** from **0 to 1** to extract data & insights effectively from bank statements.
- Engineered a **multi-model AI pipeline** (Gemini 2.5 + GPT-4) with intelligent fallback logic, improving extraction accuracy by **35%** and reducing manual review time by **44%**.

Graduate Research Assistant **Sep 2024 – May 2025**
University of Wisconsin – Madison *Madison, WI*
- Created a scalable ingestion system for **80k+** medical documents, optimizing chunking, embedding, and retrieval.
- Developed an **Agentic RAG** pipeline tailored for radiology reports, enabling clinicians to gain diagnostic insights.

Autonomous System Research Intern **Jun 2024 – Aug 2024**
Nokia Bell Labs *Murray Hill, NJ*
- Constructed a **multi-agent system** leveraging **autonomous agents** for adaptive learning & collaborative reasoning.
- Designed and built a **LangGraph**-based agentic framework with an **Orchestrator LLM Chain-of-thought** prompting for multi-step planning and optimization. Agents executed specialized tasks via **tool-calling interfaces**

Associate Engineer – AI/ML **Jul 2022 – Jul 2023**
Qualcomm *Hyderabad, India*
- Introduced evaluation frameworks and metrics for ML models to enhance efficiency and accuracy by **10.26%**.
- Benchmarked **quantized** and **pruned models** on **Snapdragon SoCs** to assess power, latency & throughput.

TECHNICAL SKILLS

GenAI Competencies: GenAI, LLMs, RAG, Multi-Agent Systems, Prompt Engineering, Vector DB, ML
Programming Languages: Python, C, C++, TypeScript, ReactJS, PyTorch, Tensorflow
Frameworks & Tools: PyTorch, TensorFlow, LangChain, LangGraph, Spark, Git, Docker, Azure, AWS, GCP, CI/CD
Libraries: NumPy, pandas, matplotlib, scikit-learn, Streamlit, NLTK, spaCy, OpenCV

SELECTED PROJECTS

Relationship Extraction using Large Language Models *Repo*
- Extracting complex relationships between entities from unstructured text data, by fine-tuning **Llama3.2 LLM**.

Optimizing Natural Language Understanding: Fine-tuning Mistral 7B *Repo*
- Fine-tuning **Mistral 7B LLM** on the Samantha dataset for conversational and contextual question answering.

SELECTED PUBLICATIONS

Palmbench: A comprehensive benchmark of compressed Large Language Models on mobile platforms
The Thirteenth International Conference on Learning Representations (ICLR), Singapore, 2025
Sign Language Translation using Multi Context Transformer.
20th *Mexican International Conference on Artificial Intelligence (MICAI), Mexico City, 2021*

EDUCATION

University of Wisconsin-Madison **Sep 2023 – May 2025**
Master of Science - Computer Sciences **CGPA: 3.827 / 4.0** *Madison, WI*
Courses: Intro to AI, Advanced NLP, Foundational Models, Big Data, Data Cleaning & Integration for Data Science.

SSN College of Engineering (Affiliated to Anna University), Chennai, India **Aug 2018 – Jun 2022**
B.Tech in Information Technology **CGPA: 8.95 / 10.0** *Chennai, India*